Zhaode Wang

AI Inference Engine Expert · On-Device LLM · High-Performance Computing

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Education

Institute of Computing Technology, Chinese Academy of Sciences State Key Laboratory of Computer Architecture	M.S. in Computer Architecture	2017 - 2020
Shandong University	B.S. in Computer Science and Technology	2013 - 2017

Experience

Alibaba · Taotian Group — **Senior Technical Expert · MNN Tech Lead** — 2020 - Present

- **MNN core architecture:** Lead MNN architecture and heterogeneous performance optimization across ARM CPUs, Metal/OpenCL/Vulkan GPUs, and Hexagon/CoreML/NNAPI accelerators, covering operator kernels, memory planning, graph optimization and fusion, and model conversion; grew MNN to 15.8K+ GitHub stars.
- **On-device AI platform:** Built an embedded Python runtime and end-to-end deployment stack, enabling dozens of algorithms for Pailitao visual search, Taobao Live, security, and feed; supported Quark, Qwen, DingTalk, Youku, Taobao Instant Commerce, and Amap.
- **MNN-LLM & production:** Initiated and currently lead MNN-LLM for 100+ language and multimodal models, spanning 2/3/4/8-bit and mixed-precision quantization, KV-cache and attention optimization, and speculative decoding; deployed on-device LLMs in Taobao and Xianyu search and recommendation at tens-of-millions-user scale.
- **On-device SLM:** Led a team to train a 0.3B SLM from scratch for on-device applications, covering data collection and quality assessment, pre-training, SFT, RL, and end-to-end model evaluation.

Cambricon Technologies — **Compiler Engineer Intern** — 2018 - 2020

- Developed compiler and linker components for Cambricon's BANG toolchain, focusing on heterogeneous linking; subsequently optimized Caffe operators for MLU accelerators.

Megvii — **Algorithm Engineer Intern** — 2019

- Contributed to traffic video understanding algorithms, focusing on lane marking detection, monocular-camera vehicle speed estimation, and traffic sign recognition.

Microsoft Asia Engineering Academy — **Software Engineer Intern** — 2016

- Evaluated LevelDB, Presto, and Hadoop for advertising data storage and analytics workloads.

Selected Publications

Efficient LLM Inference on ARM via Hardware-Aware Operator Co-Design and Heuristic Mixed-Precision Search	ISCAS	2026
RecGPT-Mobile: On-Device Large Language Models for User Intent Understanding in Taobao Feed Recommendation	SIGIR	2026
MNN-LLM: A Generic Inference Engine for Fast Large Language Model Deployment on Mobile Devices	MM Asia	2024
Walle: An End-to-End, General-Purpose, and Large-Scale Production System for Device-Cloud Collaborative Machine Learning	OSDI	2022
LISA: Locality-Informed Speculative Decoding for Accelerating Autoregressive Image Generation	ECCV	2026
ARC-Decode: Accelerated Decoding with Risk-Bounded Acceptance	ICML	2026
Prism-MoE: Efficient Dense-to-MoE Conversion for Visual Autoregressive Generation	ICML	2026
AccKV: Towards Efficient Audio-Video LLMs Inference via Adaptive-Focusing and Cross-Calibration KV Cache Optimization	AAAI	2026
VertiKV: Vertical-Integrity KV Cache Compression for Efficient Multimodal Long-Context Inference	ACM MM	2026
MadaKV: Adaptive Modality-Perception KV Cache Eviction for Efficient Multimodal Long-Context Inference	ACL	2025

Patents & Applications

BANG-Linker: Heterogeneous Linking Method, Apparatus, and Related Products BANG Simulator: Programming Debugging Method, Apparatus, and Related Products	Cambricon · Filed	2019
Solution for Deploying Large Language Models on Mobile Devices	Alibaba · Filed	2024
Saliency-Aware Blockwise Quantization via Floating-Point Zero-Point Degrees of Freedom Mixed-Precision Allocation for Low-Bit LLMs Based on Action-Conditioned Operator Error	Alibaba · Pending	2026

Honors & Competitions

IEEE AICAS Grand Challenge: LLM Software and Hardware System Co-optimization	**First Prize**	2024, 2025
NPU Competitions: Tecorigin Operator Development & SOPHGO TPU Programming	**First Prize**	2022, 2024

Skills

Languages	C, C++, Python, Assembly, CUDA, OpenCL, Vulkan, Metal, BANG, Java, Objective-C
Expertise	Compilers, AI inference engines, CPU/GPU/NPU operator optimization, model conversion, quantization, large language models